Fractions of a Group

Share the group into equal parts, then count the parts you need.

1/3 of 12: make 3 equal groups



$\frac{1}{3}$ of 12 = 4

One group out of 3.

Think division

12 ÷ 3 = 4

The bottom number tells how many groups.

2/3 of 12: two of the groups



$\frac{2}{3}$ of 12 = 8

4 + 4 = 8, two groups.

The steps

1. Divide by the bottom number: that's one group.
2. Times the top number: that many groups.

$\frac{3}{4}$ of 20: 20 ÷ 4 = 5, 3 × 5 = 15

Your turn!



$\frac{1}{4}$ of 8 = 2

For teachers

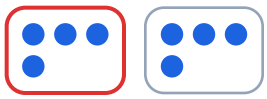
Goal: find a fraction of a group (a bridge to 4.NF.4). Say "Share it into equal groups: the bottom number says how many. One group is one third. Two groups are two thirds."

Fractions of a Group: Practice

Bottom number: how many equal groups. Top number: how many groups to count.

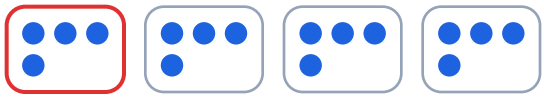
Part 1 One group The rings show the equal groups. How many in one group?

1)  $\frac{1}{3}$ of 15 = 5

2)  $\frac{1}{2}$ of 8 = 4

3)  $\frac{1}{5}$ of 20 = 4

4)  $\frac{1}{6}$ of 18 = 3

5)  $\frac{1}{4}$ of 16 = 4

6)  $\frac{1}{6}$ of 12 = 2

Part 2 Count the groups Count the dots in the marked groups.

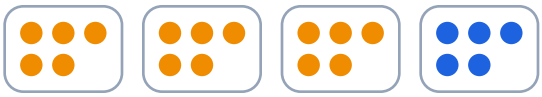
7)  $\frac{3}{4}$ of 12 = 9

Fractions of a Group: Practice (continued)

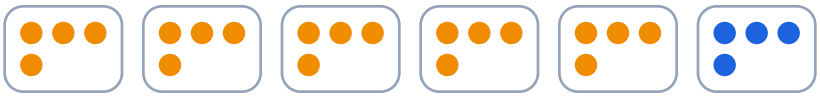
Part 2 Count the groups (continued)

8)  $\frac{2}{5}$ of 15 = 6

9)  $\frac{2}{3}$ of 18 = 12

10)  $\frac{3}{4}$ of 20 = 15

11)  $\frac{4}{5}$ of 10 = 8

12)  $\frac{5}{6}$ of 24 = 20

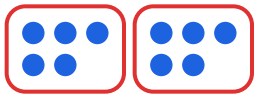
Part 3 Make the groups Circle equal groups first. The bottom number says how many.

13)  $\frac{2}{3}$ of 9 = 6

14)  $\frac{1}{4}$ of 12 = 3

Fractions of a Group: Practice (continued)

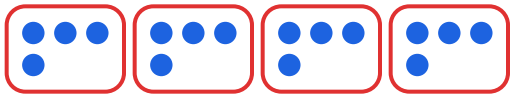
Part 3 Make the groups (continued)

15) 

$\frac{1}{2}$ of 10 = 5

16) 

$\frac{3}{5}$ of 15 = 9

17) 

$\frac{3}{4}$ of 16 = 12

18) 

$\frac{2}{7}$ of 14 = 4

Part 4 Divide, then multiply One group first, then the groups you need.

19) 24 ÷ 4 = 6 so $\frac{3}{4}$ of 24 = 18

20) 30 ÷ 5 = 6 so $\frac{2}{5}$ of 30 = 12

21) 21 ÷ 3 = 7 so $\frac{2}{3}$ of 21 = 14

22) 32 ÷ 8 = 4 so $\frac{3}{8}$ of 32 = 12

23) 36 ÷ 6 = 6 so $\frac{5}{6}$ of 36 = 30

24) 28 ÷ 4 = 7 so $\frac{3}{4}$ of 28 = 21

25) 40 ÷ 10 = 4 so $\frac{7}{10}$ of 40 = 28

26) 27 ÷ 9 = 3 so $\frac{4}{9}$ of 27 = 12

Fractions of a Group: Practice (continued)

Part 5 One group No help this time.

27) $\frac{1}{4}$ of 20 = 5

28) $\frac{1}{5}$ of 35 = 7

29) $\frac{1}{3}$ of 18 = 6

30) $\frac{1}{8}$ of 48 = 6

31) $\frac{1}{6}$ of 30 = 5

32) $\frac{1}{2}$ of 14 = 7

33) $\frac{1}{10}$ of 60 = 6

34) $\frac{1}{9}$ of 45 = 5

35) $\frac{1}{7}$ of 28 = 4

36) $\frac{1}{8}$ of 40 = 5

37) $\frac{1}{3}$ of 24 = 8

38) $\frac{1}{6}$ of 42 = 7

Part 6 More than one group No help this time.

39) $\frac{3}{5}$ of 20 = 12

40) $\frac{5}{6}$ of 24 = 20

41) $\frac{3}{4}$ of 16 = 12

42) $\frac{5}{8}$ of 40 = 25

43) $\frac{2}{3}$ of 30 = 20

44) $\frac{9}{10}$ of 50 = 45

45) $\frac{2}{9}$ of 36 = 8

46) $\frac{4}{7}$ of 35 = 20

47) $\frac{3}{4}$ of 32 = 24

48) $\frac{5}{6}$ of 18 = 15

49) $\frac{4}{5}$ of 45 = 36

50) $\frac{3}{8}$ of 56 = 21

Fractions of a Group: Practice (continued)

Part 7 Hours and dollars No help this time. 1 hour = 60 minutes, 1 dollar = 100 cents.

51) $\frac{1}{4}$ of an hour = 15 minutes

52) $\frac{1}{2}$ of a dollar = 50 cents

53) $\frac{3}{4}$ of an hour = 45 minutes

54) $\frac{1}{4}$ of a dollar = 25 cents

55) $\frac{1}{3}$ of an hour = 20 minutes

56) $\frac{1}{10}$ of a dollar = 10 cents

57) $\frac{2}{3}$ of an hour = 40 minutes

58) $\frac{3}{4}$ of a dollar = 75 cents

Part 8 Remember? Problems from earlier packets.

59) $\frac{4}{10}$ < $\frac{7}{10}$

60) $\frac{4}{10} = \frac{2}{5}$

61) $\frac{15}{3} = 5$

62) Round 917 to the nearest hundred: 900

63) 9 groups of 2 = 18

64) $\frac{1}{6}$ > $\frac{1}{10}$

Part 9 Word problems Draw the groups.

65) There are 12 kids. 1/3 of them wear hats. How many kids wear hats?

Number sentence 12 ÷ 3 = 4

Answer 4 kids

Draw or show your work

Fractions of a Group: Practice (continued)

Part 9 Word problems (continued)

66) Ana has 20 stickers. She gives away 1/4 of them. How many stickers does she give away?

Number sentence 20 ÷ 4 = 5

Answer 5 stickers

Draw or show your work

67) A box has 18 crayons. 2/3 of them are new. How many crayons are new?

Number sentence 18 ÷ 3 = 6, 2 × 6 = 12

Answer 12 crayons

Draw or show your work

68) Leo has 30 marbles. 3/5 of them are blue. How many marbles are blue?

Number sentence 30 ÷ 5 = 6, 3 × 6 = 18

Answer 18 marbles

Draw or show your work

69) Challenge: Mia read 3/4 of a 40-page book. How many pages does she have left to read?

Number sentence 40 ÷ 4 = 10, 1/4 is left

Answer 10 pages

Draw or show your work

Check-In: Fractions of a Group

Show what you know. No help this time.

Mastery: 8 of 9 (80%)

Part 1 Count the groups

1) 

$\frac{3}{5}$ of 20 = 12

Part 2 Make the groups

2) 

$\frac{2}{4}$ of 12 = 6

Part 3 One group

3) $\frac{1}{7}$ of 42 = 6

4) $\frac{1}{6}$ of 36 = 6

5) $\frac{1}{4}$ of 32 = 8

Part 4 More than one group

6) $\frac{4}{7}$ of 21 = 12

7) $\frac{2}{5}$ of 45 = 18

8) $\frac{8}{10}$ of 20 = 16

9) $\frac{7}{8}$ of 16 = 14